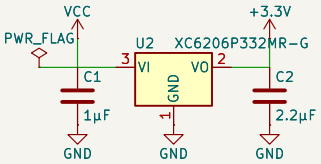
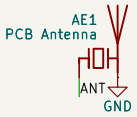
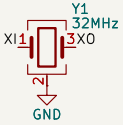


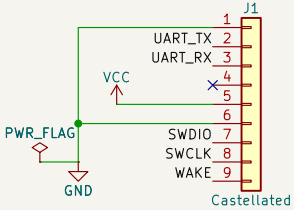
J1 castellated edge: 1.0 mm half-holes on a 0.1 in grid, three per edge.
Viewed from the top with the antenna at the bottom, pads count
counter-clockwise from the top-left. All I/O is 3.3 V.
1 GND (left edge)
2 UART_TX: PB13, module out (left edge)
3 UART_RX: PB12, module in (left edge)
4 NRST: not connected (right edge)
5 VCC in, 3.3 to 6.0 V (right edge)
6 GND (right edge)
7 SWDIO: PB14 (top edge)
8 SWCLK: PB15, also the ROM ISP strap (top edge)
9 WAKE: PA11, module out (top edge)

U2 feeds VIO33/VDD33 (pin 5). C2 2.2 uF sits at pin 5 and is also the LDO output capacitor.
DC-DC: VSW (pin 4) drives L1 10 uH into VDCID (pin 3), decoupled by C4 4.7 uF; VDCIA (pin 16) is tied to VDCID.
The DC-DC is off at power-on (VSW then passes VDD33 through L1); firmware enables it. C5 470 nF decouples VINTA (pin 14).
Y1 has no external load capacitors: the CH592 provides programmable internal load capacitance.
ANT (pin 15) drives the PCB antenna directly, as on OpenDongle.
The CH592D has no reset pin, so J1 pad 4 (NRST) is not connected. USB (PB10/PB11) is not brought out.









PB15 low at power-on puts the CH592D in ROM ISP. The boot ROM pulls PB15 up,
so leave J1 pad 8 open or high at power-up. A blank part programs over SWD (pads 7 and 8).

CERN-OHL-W CERN Open Hardware Licence Version 2 – Weakly Reciprocal
github.com/openkeyboard-org/OpenController

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Sheet: /
File: OpenController-CH592.kicad_sch

Title: OpenController CH592

Size: A4	Date: 2026-10-25	**Rev: 0.1**
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